

Price changes around hedge fund trades: Disentangling trading and disclosure effects[°]

Ettore Croci^{*}

Giovanni Petrella^{**}

This draft: March 29, 2013

Abstract

Previous studies find positive stock market reactions around announcements that hedge funds own large equity stakes in companies and use this evidence to support the hypothesis that hedge fund activism creates value. A concurrent explanation is that the price reaction reflects, at least in part, the market impact associated with hedge fund trades. We exploit the blockholder's legal filing requirements with the Italian regulatory authority (CONSOB) to separately examine trading and disclosure effects associated with hedge fund trades. Trading effects are related to hedge funds' buying activity on the market, whereas disclosure effects are related to the possibility that the market anticipates future activism and reacts to the announcement of hedge funds' ownership. The trading effect is significantly larger than the disclosure effect. This result implies that the price impact associated with hedge fund purchases explains a large portion of the price reaction attributed exclusively to hedge fund activism in previous studies. We also find that trading by hedge funds produces a permanent change in the stock price, which occurs in equal part both prior to and concurrent to the trading activity. The execution strategy used by hedge funds to accumulate their position (on exchange or downstairs trading vs. block trading or upstairs) affects both the size and the temporal distribution of the trading effect.

Keywords: hedge funds; Italian stock market; price impact; disclosure.

JEL Classification: G14; G23; G32.

[°] We are grateful to Emanuela Conti and Connie Ricciardi at Borsa Italiana for providing us with block trading data. We also thank Mike Wright, Igor Filatotchev and the participants at the conference on "Management, Governance, and Regulation in the Changing Investor Landscape" organized by Cass Business School and the Journal of Management and Governance for useful comments and suggestions.

^{*} Università Cattolica del Sacro Cuore, Milano, Italy, e-mail ettore.croci@unicatt.it.

^{**} Università Cattolica del Sacro Cuore, Milano, Italy, e-mail giovanni.petrella@unicatt.it.

I. INTRODUCTION

Hedge funds, and hedge fund activism in particular, have recently been investigated both in the U.S. (Kahan and Rock, 2007; Brav et al., 2008; Clifford, 2008; Greenwood and Schor, 2009; Klein and Zur, 2009) and in Europe (Mietzner and Schweizer, 2011; Becht et al., 2010; Weber and Zimmerman, 2011). The positive market reaction around the announcement of a large equity stake acquired by a hedge fund found in previous studies has been associated with the expectation of future activism by the new shareholder (Brav et al., 2008; Klein and Zur, 2009). Hedge funds are in fact considered more effective monitors than other institutional investors because of the strongest incentives of their managers both in terms of direct compensation from the fund and in terms of personal wealth invested in the fund. In addition, hedge funds have fewer conflicts of interests than other institutional investors, such as pension funds and mutual funds, that often have other business relations with the target company or non-financial agendas and goals (Brav et al., 2009). Additionally, hedge funds are less constrained by regulation than pension and mutual funds. Overall, the literature indicates that hedge funds create value for other shareholders not just because they are good stock pickers, but also because of their activism (Brav et al., 2008).

However, while activism-related explanations have found support in the literature, a concurrent—but not mutually exclusive—explanation is that the positive price reaction documented in previous studies reflects, at least in part, the market impact associated with hedge fund trades, and not just the expectation of their activism. In this paper, we separate the trading and disclosure effects to analyze whether the stock price reaction around the announcement of hedge fund shareholding is due to the anticipation of future activism or to a volume (or price pressure or market impact) effect. Trading effects are related to the hedge funds' buying activity, whereas disclosure effects are related to the possibility that the market reacts to the announcement of hedge funds' ownership in anticipation of their activism. Trading effects cannot be associated with the anticipation of the hedge funds' activism because the identity of the investor that is acquiring the equity stake is not publicly disclosed at the time of the trade. Both trading and disclosure effects are further decomposed into a permanent component (i.e., information-related) and a transitory component (i.e., liquidity-related).

Italy is an ideal venue to address this research question for two reasons. First, the Italian regulation requires any investor with a beneficial ownership of 2% or more to disclose his/her holding as well as the date of trading. This enables us to separately examine trading effects and disclosure effects associated with hedge fund ownership based on the data provided by the Italian regulatory authority (CONSOB). Second, Italy has a very low threshold that triggers public disclosure (2%) compared with the international financial market landscape. For example, the disclosure threshold is 5% in the U.S. The examination of a country with a low threshold is important because it allows us to draw inferences on whether or not setting a low threshold is beneficial to minority investors. In fact, the difference between a disclosure threshold at 5%, such as in the U.S., and one at 2% such as in Italy affects both the trading effect and the disclosure effect. In the case of the former, hedge funds purchase a larger fraction of the firm's equity and consequently generate more pressure on the company's stock price; in the latter, a 5% equity stake acts as a stronger signal than a 2% stake regarding the future involvement of the hedge fund in the company.

Four main findings arise from our analysis. First, the trading effect is significantly larger than the disclosure effect. This implies that the price impact associated with hedge fund purchases on the market explains a large portion of the price reaction that was attributed exclusively to hedge fund activism in previous studies. This finding has the following important implication: it signals that the returns around the announcement of a hedge fund's filing may exaggerate the true effect of future activism, especially when the two effects cannot be disentangled, such as in the U.S. setting. Second, we find that trading by hedge funds produces a permanent change in the stock price, which occurs in equal part prior to and concurrent with trading. This means that hedge fund trading activity affects stock prices on its own, regardless of the information effect associated with the hedge fund ownership, which in our research design occurs later in the timeline of events. Third, the execution strategy used by hedge funds to accumulate their position (on exchange or downstairs trading vs. block trading or upstairs) affects the size of the estimated trading effect: the overall price change associated with the trading date is much larger for downstairs trades than for upstairs trades. Fourth, the execution strategy also affects the temporal distribution of the overall trading effect: for downstairs trades, a large portion of the overall price change occurs prior to the trading date filed with the

market authority whereas, for upstairs trades, the overall price change accumulates on the trading date filed with the market authority.

This study offers several contributions to the literature. First, we decompose price changes around hedge fund trades in trading and disclosure effects. Previous studies were not able to distinguish the sources of the price reaction and attributed the entire reaction to the disclosure effect. We show that the price reaction around the announcement is not entirely attributable to the expectation of activism. As the literature largely ignores the trading effect, our results indicate that the (initial) return of activism may be overstated because it includes both the “pure” activism effect and the trading effect. Second, we are able to differentiate the trading effects associated with different execution strategies used by the hedge fund to build its position (i.e., block trading vs. on-exchange trading). Third, we also study the behavior of trading volume around hedge fund trades. This allows us a more comprehensive examination of the effects of hedge fund trading on equity markets. An examination of the price and volume patterns around hedge fund trades provides insights relevant to understanding the impact on market conditions of this category of investors who is responsible for about 30 % of U.S. equity markets trading volume (Blundell Wignall, 2007).¹ Fourth, we use a clean sample to estimate the price reaction associated with uncontaminated events and show that results are highly sensitive to the inclusion of contaminated events.² In fact, hedge fund trades may also occur in conjunction with other events affecting corporate cash flows, which might contaminate the market reaction around the announcement. Finally, we discuss the costs and benefits of a regulation with low disclosure thresholds.

The remainder of this study is organized as follows. Section 2 provides a concise literature review about the effects of hedge funds trading and states the hypotheses that will be tested in the empirical analysis. Section 3 describes the institutional details of the Italian equity market, the sample selection procedure, and the data used in the empirical

¹ In testimony before the U.S. Senate Judiciary Committee, the Director of Enforcement at the S.E.C., Linda Thomsen (2006), stated that “insider trading by hedge funds remains a substantial concern to the Division, and represents a significant focus of our current enforcement efforts” and also confirmed that the Division was closely monitoring hedge fund trading activity.

² We report the results from both the full sample and the uncontaminated sample to provide evidence of the effects of using unclean observations.

analysis. Section 4 discusses methodology and empirical findings. Summary and conclusions are provided in the last section.

II. THE EFFECTS OF HEDGE FUND PURCHASES: LITERATURE REVIEW AND HYPOTHESES DEVELOPMENT

Previous studies on the impact of shareholder activism on target companies show negligible impacts on share values (see, for example, Karpoff, 2001; Gillan and Starks, 2007; and Ferri, 2011, for excellent surveys of empirical studies). By contrast, a few recent papers studying hedge fund investment activity find that hedge fund activism generates statistically significant abnormal returns and conclude that hedge fund intervention is beneficial to all shareholders (Brav et al., 2008; Klein and Zur, 2009).

These different results are usually ascribed to the unique characteristics of hedge funds compared with other institutional investors (Brav et al., 2009). Hedge fund managers have greater incentives in monitoring CEOs and controlling shareholders of the target companies because their compensation depends on the performance of the fund. Unlike mutual fund and pension fund managers, hedge fund managers often have a substantial part of their wealth invested in the fund. Hedge funds are not obliged to maintain diversified portfolios or to adhere to strict fiduciary standards such as, for example, those set by the Employee Retirement Income Security Act (ERISA) in the U.S.

Overall, hedge funds are lightly regulated compared with other investment vehicles, although in recent years there has been an attempt to increase hedge fund regulation. For example, the Private Fund Investment Advisers Registration Act, a sub-part of the Dodd-Frank Act of 2010, implemented new requirements on hedge funds and their advisers in the U.S. and increased supervision by the Securities and Exchange Commission (SEC) on hedge fund activity. To increase transparency, hedge funds are now required to register with the SEC.

Hedge funds investors cannot exit their investments for a minimum period of time (lock-up agreements). Additionally, hedge funds are relatively free from the types of conflicts of interest that afflict other institutional investors. In fact, they usually do not have business relationships with the firms they target like mutual funds (Davis and Kim,

2007), and their managers rarely have a political agenda such as those in some public pension funds (Romano, 1993; Hebb and Beeferman, 2009) or union funds (Partnoy and Thomas, 2007). These characteristics favor activism and monitoring by hedge funds.

However, hedge funds also have a dark side. Schneider and Ryan (2011) suggest that although hedge fund activism generally benefits other shareholders, the interests of some hedge funds might occasionally be contrary to the interest of other investors and generate a “principal-principal” conflict. The authors consider short selling strategies, involvement in bankruptcy, and the use of leverage as sources of inter-principal conflicts.

Brav et al. (2008), Klein and Zur (2009), and Boyson and Mooradian (2011) study hedge fund activism in the U.S. and find that—on an ex-ante basis—the stock market reacts favorably to the announcement of hedge fund interventions and—on an ex-post basis—target firms experience an improvement in operating performance after the intervention. Klein and Zur (2009) also compare hedge fund activist campaigns with other private investors’ activist campaigns, finding that hedge fund purchases generate larger market reactions than those associated with other activists.³ These findings, which are broadly consistent with those reported by Brav et al. (2008), suggest that the market believes, on average, that activism creates shareholder value and that the market reacts more favorably to hedge fund activism than to other private investor activism.

As a possible explanation for these findings, Greenwood and Schor (2009) observe that the abnormal returns associated with hedge fund target firms are partially explained by the ability of the hedge funds to put the company “in play” (i.e., to become a potential target in an acquisition). In fact, they document that firms targeted by activist hedge funds are more likely to be taken over than control firms. In their sample, activists increase the likelihood of a takeover by 11%.

Positive excess returns might also be explained by the market anticipating that hedge fund-monitoring activity reduces agency costs of targeted firms or simply that hedge funds select better investments than non-hedge fund investors. Clifford (2008) provides evidence to support the first view. In fact, he finds that the market views a block

³ The group of non-hedge funds activists includes individuals, private equity funds, venture capital firms, and asset management groups.

ownership by a hedge fund as a positive signal regardless of its intent and also that the market views shareholder activism as providing additional value to the target firm. Comparing active and passive hedge funds, Clifford finds that firms targeted by both experience positive and significant excess returns surrounding the filing date.⁴ However, firms targeted by activist hedge funds earn significantly higher returns, confirming the general conclusion that activism is beneficial to all shareholders.

While the hedge fund-activism literature has mainly focused on the U.S. market, there are a few studies concerning Europe (Mietzner and Schweizer, 2011; Becht et al., 2010; Weber and Zimmerman, 2011).⁵ In their cross-countries study, Becht et al. (2010) examine hedge fund activism in Europe using both public and private hedge fund interventions. They find that, similar to the U.S. literature, the public announcement of a hedge fund purchase generates an increase of 4.4% in the stock price of the target company in the event window [-20, +20], but this reaction varies significantly depending on the country of incorporation. In fact, the largest market reactions for the event window [-20, +20] are observed in Germany (6.0%), while the abnormal return is between 2.6% and 2.8% in countries like Italy, France, and the U.K. There are a few studies that examine hedge fund activism in Germany, where hedge fund and private equity deals have been under scrutiny by the press and politicians. Mietzner and Schweizer (2011) find that the announcement of a hedge fund purchase in Germany is welcome by the market (6.2% in the [-20, +20] event window), but different from the U.S. literature, this positive reaction is not due to a reduction of agency costs. In fact, Mietzner and Schweizer (2011) find that abnormal volume positively affects the cumulative abnormal returns (CARs) around a hedge fund announcement, which is consistent with the explanation that buy-side pressure drives positive CARs. Weber and Zimmerman (2011) take a closer look at the impact of hedge fund trading to disentangle information effects from price-pressure effects. They find that the price effect around the date on which the hedge fund buys and sells the stock of the target company, which

⁴ An active investor who seeks control at the target company or plans to influence the firm management needs to file a Schedule 13D. By contrast, passive institutional investors that acquire more than 5%, but less than 10%, of a company's stock and do not intend to seek to influence control at the target company, but are merely investing, file a Schedule 13G.

⁵ Europe and the U.S. are not the only countries examined in this area of research. For example, Hamao et al. (2010) investigate hedge fund activism in Japan.

cannot be due to information because the information has not been disclosed to the market yet, is at least as important as the information effect around the publication date (0.88%). They also collect evidence suggesting the possibility of insider trading around the announcement date.

Table 1 reports the cumulative abnormal returns estimated by the previous studies on hedge fund interventions. This table shows heterogeneity in the short-term value effects associated with hedge fund interventions, which vary from less than 1% to about 8%. This variation mostly depends on the width of the event window used in these studies, which can be more or less affected by trading. In fact, Klein and Zur (2009) explicitly motivate the use of event windows beginning on day -30 “to allow for the ten-day 13D filing window, possible prior leakage of information, and pre-filing price pressure that may occur due to the activist accruing a large stake in a relatively short period of time”.⁶ Our paper draws on this mixture of factors and aims at separating the information-related and price-pressure components.

Trading volume investigation can also help us in understanding the stock price reaction. Brav et al. (2008) study the behavior of trading volume around the filing of a Schedule 13D and find that the spike in the abnormal trading volume “*does not occur on the event day but rather during the ten-day period before the filing*”.⁷ This evidence is clearly consistent with the fact that investors are required to file a Schedule 13D within ten days after the transaction that causes them to exceed the 5% threshold and it is therefore possible that the filing hedge fund was already buying shares in the days immediately prior to the filing date.⁸ Consequently, the 3.2% run-up prior to the filing date that they find might be explained by the buying activity of the hedge fund.⁹ Our paper focuses exactly on this point and tries to disentangle disclosure and trading effects.

⁶ See Klein and Zur (2009, page 26).

⁷ See Brav et al. (2008, page 1756).

⁸ In some cases hedge fund intervention is first announced in a news story published prior to the 13D filing.

⁹ Mietzner and Schweizer (2011) show that evidence from the German market is also consistent with the explanation that positive CARs are driven by the hedge fund trading activity.

Based on the previous discussion, we test the following hypotheses:

Hypothesis 1a (Information effect): *If the reaction on stock prices and volumes around the hedge fund's announcement is due to an information effect and the information is deemed to be positive, then this reaction should be positive and concentrated around the disclosure date.*

Hypothesis 1b (Trading effect): *If the reaction on stock prices and volumes around the hedge fund's announcement is due to a trading effect, then the reaction should be positive and concentrated around the trading date.*

Exploiting the fact that before the entry into force of the MiFID regulation (i.e., before November 1, 2007) investment firms were required to disclose block trades to the Italian Stock Exchange, we are also able to test whether and how the execution strategy chosen by the hedge funds impacts the distribution of the abnormal returns around the trading date.¹⁰ The price change associated with downstairs trading is related to the trading activity of the hedge fund, whereas the price effect of the upstairs trading (or block trading) is simply due to the market reaction associated with the disclosure of a transaction that took place off exchange.

Hypothesis 2 (Downstairs vs. upstairs): *Abnormal returns observed prior to the trading date are larger for downstairs trades than for upstairs trades.*

Block transactions can be buyer initiated or seller initiated: in the former the block price is greater than the current market price, in the latter the block price is lower than the current market price. If the hedge fund buys through a seller-initiated block trade, it pays a price lower than the market price because of its passive trading strategy. If, by contrast, the hedge fund buys through a buyer-initiated block trade, it will pay a premium to the seller to acquire the shares. Consequently, the composition of the trades (i.e., buy blocks vs. sell blocks) affects the size of the trading effect observed for upstairs trading.

¹⁰ Under MiFID regulation, from November 1, 2007 onwards, investment firms are requested to send information about block trades to the relevant market authority (i.e., to the CONSOB for shares listed in the Italian market) and no longer to the Italian Stock Exchange.

Additionally, Kraus and Stoll (1972), Chan and Lakonishok (1993), and Keim and Madhavan (1996) observe asymmetric returns after both block purchases and block sales. This leads to our third hypothesis:

Hypothesis 3 (Buy blocks vs. sell blocks): *Abnormal returns observed for buy blocks are larger than those observed for sell blocks.*

III. DATA AND SAMPLE DESCRIPTION

We use the CONSOB shareholding database to construct our sample of hedge fund purchases in the Italian stock market.¹¹ CONSOB is the public authority responsible for supervising and regulating the Italian securities market. Its shareholding database contains all the filings required by the Italian regulation concerning blockholdings in shares of a listed company. Specifically, for each blockholding, the database contains the disclosure date, the trading date, the percentage of the company's equity acquired, and the percentage owned after the trade.

The Italian regulation requires that an investor must disclose any ownership of more than 2% of a publicly listed company's equity within 5 trading days after the acquisition. This threshold is significantly lower than the 5% level required in the U.S. and in most of the EU countries to trigger a public notification of an investor's holding in a listed company.¹²

We use a list of more than a hundred well-known hedge funds to search for the hedge fund name in the CONSOB database. We relied on several sources to compile this list: we used funds' Internet sites, investor magazines, newspapers, and other sources to determine if the filing investor was a hedge fund at the time of the deal.¹³ We consider CONSOB filings between January 1, 2000 and December 31, 2007, and we restrict our attention to the initial shareholding of a hedge fund in a target company. This restriction is aimed at capturing the first news concerning the hedge fund's shareholding in the

¹¹ The CONSOB database is available online at http://www.consob.it/main/en/issuers/listed_companies/advanced_search/index.html.

¹² The disclosure threshold for a blockholding in a listed company varies across Europe. The threshold is 3% in the U.K. and Germany, and 5% in the other countries. Italy has the lowest threshold.

¹³ The list of hedge funds is available from the authors.

company. While later filings allow the market to update its beliefs on the deal, the news about the initial shareholding enables us to examine the market reaction to the arrival of the hedge fund in the ownership structure of the firm. Our search produced 193 pairs of hedge funds and target firms. As we are interested in studying the market reaction to the hedge fund acquisition, we exclude from the sample 32 observations in which the purchase took place at or immediately after the IPO of the target company. Our final sample consists of 161 hedge fund-target pairs, involving 38 hedge funds, and 123 unique target companies.

We filter out the initial sample (hereafter *full sample*) to exclude confounding effects. In particular, we want to make sure that the event under examination is not contaminated by the disclosure of other news that might have had an impact on the stock price of the target firm but are unrelated to the hedge fund purchase. To clean our sample, we searched the database of *Il Sole-24 Ore*, the leading Italian financial newspaper, for company news announced in the two-week periods centered around both the trading date and also the announcement date. We found 35 hedge funds purchases to be contaminated by other events that might undermine the assumption that the market reaction is exclusively due to the hedge fund acquisition. The remaining 126 hedge fund-target firm pairs constitute our final sample (hereafter *Uncontaminated sample*).

Hedge funds may accumulate their position in a target firm by acquiring shares in the exchange or off the exchange (i.e., in the upstairs or block market). Consequently, the trading effect associated with hedge fund purchases may differ according to the hedge fund's execution strategy. To control for this factor, our dataset also includes information—obtained from the Italian Stock Exchange (Borsa Italiana)—about trades executed in the upstairs market.

Table 2 reports the number of hedge fund-target company pairs partitioned by year (Panel A) and hedge fund (Panel B) for each of the two samples (Full and Uncontaminated). First, our identification procedure allows us to discover a fairly large sample of hedge fund acquisitions in Italy. We are able to identify 161 initial announcements of hedge fund stakes. The sample size is larger than the one in Weber and Zimmerman (2011) and Mietzner and Schweizer (2011) for Germany, and almost six times as large as the Italian sample used by Becht et al. (2010), which counts 28

observations only. While Becht et al. (2010) created their sample using keyword searches for press articles on the Factiva database, we use the CONSOB database to identify all the hedge fund shareholdings in our sample period. As expected and similar to Becht et al. (2010) and Erede (2009), no hedge fund is located Italy—most are domiciled in the U.S., U.K., or the Caribbean Islands. We also report the average stake (i.e., the percentage of shares) acquired in each firm. The hedge funds more active in the Italian market in the 2000–2007 period are Amber (23 stakes acquired), Egerton (19 stakes acquired), and Gartmore (14 stakes acquired). The average stake acquired is 3.1%. This low initial average stake is not surprising because the threshold that triggers a public disclosure in Italy is only 2%. Looking at the time series, we note that the average stakes are higher in 2000 and 2007 (around 4.4%), and less than 3% in the other years except 2001. Similar to the U.S. evidence (Brav et al., 2009), most of the acquisitions took place from 2005 to 2007, the period immediately before the financial crisis.

IV. EMPIRICAL ANALYSIS

IV. A. Methodology

We examine the behavior of the target company's stock price around two event dates: the hedge fund trading date and the hedge fund disclosure date. We compute the overall price change over the window $(-4, 3)$, where 0 is the event date (trading or disclosure), and decompose it into three parts. First, a pre-event component that accounts for, respectively, the market impact of purchases prior to the trading date and the possible leakage of information prior to the disclosure date. This component refers to the window $(-4, -1)$. The second part is the event component that is computed over the window $(0, 1)$. This component refers to the market reaction to the announcement when it is computed around the disclosure date, and to the market impact of purchases when it is computed around the trading date. Third, a post-event component that accounts for transitory price effects (or price reversals) is computed over the window $(2, 3)$. The sum of the three components is the overall price change and represents the permanent price impact (PPI) associated with the event under investigation (either trading or disclosure). Figure 1 illustrates a graphical representation of the decomposition method (Part 1a for the disclosure event and Part 1b for the trading event).

Our measure of price change controls for systematic risk. We estimate market model parameters over the period (-240, -41) and compute beta-adjusted returns for each target company.

To perform the event study on the firm's trading volume, we follow a market model approach like Sanders and Zdanovic (1992).¹⁴ As in Ajinkya and Jain (1989), we do not perform the event study on the raw trading volume, which is highly non-normal, but on the transformed variable $v_{it} = (1 + \ln \text{volume}_{it})$. The trading volume of the local market index is used in the market model regressions. As we did for the target stock returns, we estimate market model parameters for trading volume over the period (-240, -41).

IV.B. Effects on price behavior

Table 3 reports the results for the event studies performed on target company stock returns and centered around the trading date (Panel A) and the disclosure date (Panel B) for each of the two samples (Full and Uncontaminated).

The overall price reaction associated with a hedge fund trade is much larger than that associated with the disclosure event: 1.76% vs. 0.93%. As for the trading effect, we do not find any price reversal after the initial market reaction. The overall price change, which represents the permanent price impact of the trade, is equally split between the pre-event period (0.88%) and the event date (0.87%). The magnitude of the price reaction associated with hedge fund trades decreases as long as we eliminate contaminated events from the dataset—the total price change drops from 1.76% to 1.37%. Every component of the total price change drops when using the uncontaminated sample. As for the disclosure effect, we do not find any evidence of leakage: the price change occurring in the pre-event period is not statistically significant. The overall effect associated with the disclosure of hedge fund purchases only comes from the market reaction to the news observed in the event period (0,1). Interestingly, the price reaction is not statistically significant in the uncontaminated sample. Our evidence is consistent with the hypothesis that most of the price effect observed in previous studies in conjunction with hedge fund trading was attributable to a price-pressure effect associated with trading. As Table 1 clearly shows that hedge funds usually declare

¹⁴ However, we do not include dummies for the day-of-the-week effect.

ownership stakes of little more than 2% (and well below the 5% threshold adopted in the U.S.), our results also raise some doubts about the fact that a mere 2% ownership stake is enough to create the expectation of future activism by the hedge fund.

The execution strategy used by hedge funds to accumulate their position significantly affects the estimated trading effect. Table 4 reports for each of the two samples target company abnormal stock returns centered around the trading date and partitioned by trading venue: downstairs market (or on-exchange trading) in Panel A and upstairs market (or block trading) in Panel B.

First, we find that the overall price change associated with the trading date is larger for downstairs trades (2.29%) than for upstairs trades (1.04%), although the difference is not statistically significant. The former require the hedge fund buying shares through the electronic trading system of the Italian exchange; the latter imply a negotiated deal in the block market. The market price reacts more strongly when the hedge fund buys on the exchange than when it buys from a dealer off market. This is again consistent with a trading pressure explanation of the market reaction.

Second, looking at the decomposition of the overall price change, we observe the following pattern: for downstairs trades, a large portion of the overall price change occurs in the four-day window prior to the trading date filed with the market authority; whereas, for upstairs trades, the total price change almost completely accumulates over the (0, 1) event window, where 0 is the trading date filed to the market authority. Indeed, the abnormal price reaction in the window (0,1) accounts for about 100% of the total price change for upstairs trades, while it accounts for only 30% for downstairs hedge fund trades. Consequently, the price change observed in the window (-4, -1) accounts for the remaining 70% of the total trading effect for downstairs trades. This evidence is consistent with our expectations: for downstairs trades, the hedge fund buying activity drives the price upward during the days immediately preceding day 0 when the hedge fund buys the shares through on-exchange transactions, whereas for off-exchange transactions the stock price does not react until, as required by regulation, the upstairs trade has to be disclosed to the market as a block trade (prior to the disclosure as a block

holding).¹⁵ This implies that for upstairs trades the market reacts to the announcement of the block trade on the event date. Consequently, in the interpretation of the empirical findings, we need to consider that the price change associated with downstairs trading is related to the trading activity of the hedge fund, whereas the price effect of the upstairs trading is the market reaction to the disclosure of a transaction that took place off exchange.

The portion of price change occurring in the pre-event period (-4, -1) is statistically insignificant and economically negligible (-0.11%) for upstairs trading, and statistically significant and economically important for downstairs trading (1.61%). Such a large price change, which accounts for 70% of the total price change, may be explained by the order imbalance that may occur prior to day 0 due to the hedge fund accruing a large stake using on-exchange trades. Conversely, we do not find any evidence of statistically significant price changes in the pre-event period for upstairs trades. This result is consistent with the explanation that for such trades there is no order imbalance that takes place in the exchange because these are by definition off-exchange transactions. This finding also offers an indirect confirmation of the fact that the price effect associated with hedge fund trading is mostly due to the market impact of the hedge fund trading activity. In short, our conclusion is that the execution strategy affects the temporal distribution of the overall trading effect.

To study whether the trading effect observed for upstairs trades depends upon the composition of the trades, we partition the subsample of upstairs trades in buy blocks and sell blocks. We classify buy and sell blocks according to the party who initiates the trade. Table 5 reports target company abnormal stock returns for upstairs trades only, centered on the trading date, and partitioned by trade direction: buy blocks in Panel A and sell blocks in Panel B. We find that the price reaction for buy blocks is much larger than that associated with sell blocks: 1.93% vs. 0.49%. This evidence is consistent with Kraus and Stoll (1972), Chan and Lakonishok (1993), and Keim and Madhavan (1996) who observe positive (and asymmetric) returns after both block purchases and block sales. We also find that the overall price change almost entirely accumulates over the (0, 1) event window. The fact that there is no leakage effect going on is not surprising

¹⁵ In fact, the Italian regulation requires that block trades have to be disclosed within just one hour from the execution of the trade.

because as an off-exchange trade there is no trading pressure on the price prior to the announcement. We also do not find evidence of reversal. In short, given that the block trade was disclosed within one hour from the trade, the observed price change only reflects the market reaction to the news about the block.

In a multivariate framework, we estimate cross sectional regressions to investigate main factors affecting the price change associated with hedge fund trading and the market reaction to the news of hedge fund ownership. We are interested in cross checking the previous evidence about the importance of the trading effect: if trading is important, then the size of the trading activity by the hedge fund should affect the size of the price reaction observed on the market. Table 6 presents the estimates of OLS regression models where the dependent variable is the CAR in the event window $(-4, +3)$ in Panel A, the CAR in event window $(-4, -1)$ in Panel B, and the CAR in the event window $(0, 1)$ in Panel C on a constant, the percentage of target's equity disclosed, a dummy for upstairs trades (Block) and a dummy for buy blocks (Buy Block). Models I–III explain the CAR centered on the trading date, and models IV–VI refer to the CAR centered on the disclosure date.

The only statistically significant variable is the percentage of equity acquired by the hedge fund. This variable contributes to the explanation of the pre-event (i.e., CAR $(-4, -1)$) and the event date (i.e., CAR $(0, 1)$) price change associated with the trading date. No variable is statistically significant in the specifications estimated for the CAR centered on the disclosure date. This finding is consistent with the idea that the trading pressure exerted by the hedge fund when acquiring the stake directly affects the observed price change: the larger this stake, the larger the trading pressure, and the larger the observed price change. This evidence provides additional confirmation to our results about the importance of the trading effect.

IV.C. Effects on volume behavior

We perform a volume event study on the effects of hedge fund trading on the overall market trading volume. Table 7 and Fig. 2a report the estimates of the daily abnormal trading volumes in the eight-day period $(-4, +3)$ around trading and disclosure dates for the Full Sample (Panel A) and the Uncontaminated Sample (Panel B). We find evidence

of abnormal trading activity around the trading date and, as expected, no evidence of abnormal trading activity around the disclosure event.

Figure 2b shows that the temporal distribution of the abnormal trading activity around the trading date depends on the trading venue: abnormal volumes rise on the trading date when the hedge fund uses the upstairs market, whereas abnormal trading activity is evident prior to the disclosed trading date when the hedge fund uses the downstairs market. This latter evidence is the result of the hedge fund buying activity on the stock exchange, whereas the former evidence is the market reaction stimulated by the disclosure of the block trading.

V. SUMMARY AND CONCLUSIONS

This paper examines whether the observed price changes around the disclosure of hedge fund ownership stakes are due to the expectation of future activism or to a trading effect. Previous studies find positive market reactions around the filing of an SEC Schedule 13D (a form to be filled out by active investors) and, once the hypothesis that hedge funds are merely good stock-pickers is discarded, they use this evidence to support the hypothesis that hedge fund activism creates value. We argue and show in this paper that factors other than the expectation of future activism may concur to generate this market reaction. Specifically, the price reaction also reflects the market impact associated with hedge fund trades.

We exploit the blockholder's legal filing requirements with the Italian regulatory authority (CONSOB) to separately examine trading effects and disclosure effects associated with hedge fund trades. Trading effects are related to the hedge funds buying activity, whereas disclosure effects are related to the possibility that the market reacts to the announcement of hedge funds ownership. Both trading and disclosure effects are further decomposed into a permanent (i.e., information-related) component and a transitory (i.e., liquidity-related) component.

We document that the trading effect is significantly larger than the disclosure effect. Our study shows that the price impact associated with hedge fund purchases explains a consistent portion of the price reaction that previous studies attribute to hedge fund

activism, whose estimated effect can therefore be inflated by the ignored trading effect. The change in the stock price generated by hedge fund trading is permanent, and it occurs in equal part prior to the trading and concurrent to the trading.

We also investigate the execution strategy used by hedge funds to accumulate their position (on-exchange or downstairs trading vs. block trading or upstairs). We find that the execution strategy affects the magnitude of the trading effect. In fact, the overall price change associated with the trading date is more pronounced for downstairs trades than for upstairs trades. Finally, the execution strategy also affects the temporal distribution of the overall trading effect. In downstairs trades, a large portion of the overall price change occurs prior to the trading date filed with the market authority whereas in upstairs trades, the overall price change accumulates over the trading date filed with the market authority.

A caveat is in order when we compare the magnitude of the abnormal returns around the announcement of the hedge fund trades in Italy with those observed in previous papers. The abnormal returns that we find in our sample are clearly smaller than what it is usually found in the U.S. and in part of the European market. While a smaller expectation of future activism can certainly explain the lower abnormal returns that we observe in the Italian market, we believe that the low disclosure threshold enforced in Italy (2%) also explains this result.

Our results have two implications. First, in the debate about hedge funds, our findings show that the price reaction observed in conjunction with hedge fund ownership—and previously only associated with the activism of hedge funds—is actually also the result of the trading activity carried out by the hedge fund on the market to acquire the equity stake. Second, on a more general level, our study shows that any time we associate a wealth (i.e., a value creation) effect to the disclosure of ownership by a specific category of investors (e.g., hedge funds, private equity firms, sovereign wealth funds, and so on) we should also carefully control for the change in price due to the way in which the investor accumulates the position (i.e., to the market impact of his/her trading)

REFERENCES

- Ajinkya, B. B., Jain, P. (1989). The behavior of daily stock market trading volume. *Journal of Accounting & Economics*, 11, 331–359.
- Becht, M., Franks, J., Grant, J. (2010). Hedge Fund Activism in Europe, *ECGI - Finance Working Paper No. 283/2010*.
- Blundell Wignall, A. (2007). An Overview of Hedge Funds and Structured Products: Issues in Leverage and Risk. *Financial Market Trends*, 92(1) 37–57.
- Boyson, N., Mooradian, R. (2011). Corporate governance and hedge fund activism. *Review of Derivatives Research*, 14, 169–204.
- Brav, A., Jiang, W., Partnoy, F., Thomas, R. (2008). Hedge Fund Activism, Corporate Governance, and Firm Performance. *Journal of Finance* 63(4), 1729-1775.
- Brav A., Jiang W., Kim H. (2009), “Hedge Fund Activism: A Review”, *Foundations and Trends in Finance* 4, 185-249.
- Chan, L., Lakonishok, J. (1993), “Institutional Trades and Intra-Day Stock Price Behaviour”, *Journal of Financial Economics* 33, 2, 173-199.
- Clifford, C. (2008), “Value creation or destruction? Hedge funds as shareholder activists”, *Journal of Corporate Finance* 14, 4, 323-336.
- Davis G.F., Kim, E.H. (2007) How do Business Ties Influence Proxy Voting by Mutual Funds? *Journal of Financial Economics*, 85, 552–570.
- Erede, M. (2009). Governing Corporations with Concentrated Ownership Structure: Can Hedge Funds Activism Play Any Role in Italy? *CLEA 2009 Annual Meeting Paper*.
- Ferri, F. (2010). Low-Cost’ Shareholder Activism: A Review of the Evidence. *Working paper*.
- Gillan, S.L., Starks, L. (2007). The Evolution of Shareholder Activism in the United States. *Journal of Applied Corporate Finance*, 19, 55–73.
- Greenwood, R. and Schor, M. (2009). Investor activism and takeovers. *Journal of Financial Economics*, 92, 362–375.
- Hebb T., Beeferman L. (2009). US Pension Funds’ Labour-Friendly Investments. *Journal of Comparative Social Welfare*, 25 (2), 109-117.
- Hamao, Y., Kutsuna, K., Matos, P.P. (2010). U.S.-Style Investor Activism in Japan: The First Ten Years. *Marshall School of Business Working Paper No. FBE 06-10; ECGI - Finance Working Paper No. 290/2010; 5th Annual Conference on Empirical Legal Studies Paper*.
- Kahan, M., Rock E.B. (2007). Hedge Funds in Corporate Governance and Corporate Control. *University of Pennsylvania Law Review*, 155, pp. 1021–1093.
- Karpoff, J. (2001). The Impact of Shareholder Activism on Target Companies: A Survey of Empirical Findings, Working Paper, University of Washington.

- Keim, D., Madhavan, A. (1996). The Upstairs Market for Large-Block Transactions: Analysis and Measurement of Price Effects. *Review of Financial Studies*, 9(1), 1–36.
- Klein, A., Zur, E. (2009). Entrepreneurial Shareholder Activism: Hedge Funds and Other Private Investors. *Journal of Finance*, 64(1), 187–229.
- Kraus, A., Stoll, H. (1972). Price Impacts of Block Trading on the New York Stock Exchange. *Journal of Finance*, 27(3), 569–588.
- Mietzner, M., Schweizer, D. (2011). Hedge Fund Versus Private Equity Funds as Shareholder Activists — Differences in Value Creation. *Journal of Economics and Finance*, forthcoming, <http://dx.doi.org/10.1007/s12197-011-9203-x>.
- Partnoy F.S., Thomas R.S. (2007). Gap Filling, Hedge Funds, and Financial Innovation. *New Financial Instruments and Institutions*, Brooking Institution Press.
- Romano R. (1993). Public Pension Fund Activism in Corporate Governance Reconsidered. *Columbia Law Review*, 93, pp. 795–853.
- Sanders, R., Zdanowicz, J. (1992). Target Firm Abnormal Returns and Trading Volume Around the Initiation of Change in Control Transactions. *Journal of Financial and Quantitative Analysis*, 27, 1, 109–129.
- Schneider, M., Ryan, L. V. (2011). A review of hedge funds and their investor activism: Do they help or hurt other equity investors? *Journal of Management and Governance*, 15(3), 349–374.
- Thomsen, L. (2006). Examining Enforcement of Criminal Insider Trading and Hedge Fund Activity. Testimony before the Senate Committee on the Judiciary, December 5.
- Weber, P., Zimmermann, H. (2011). Hedge Fund Activism and Information Disclosure: The Case of Germany. *European Financial Management*, forthcoming. doi: 10.1111/j.1468-036X.2011.00626.x

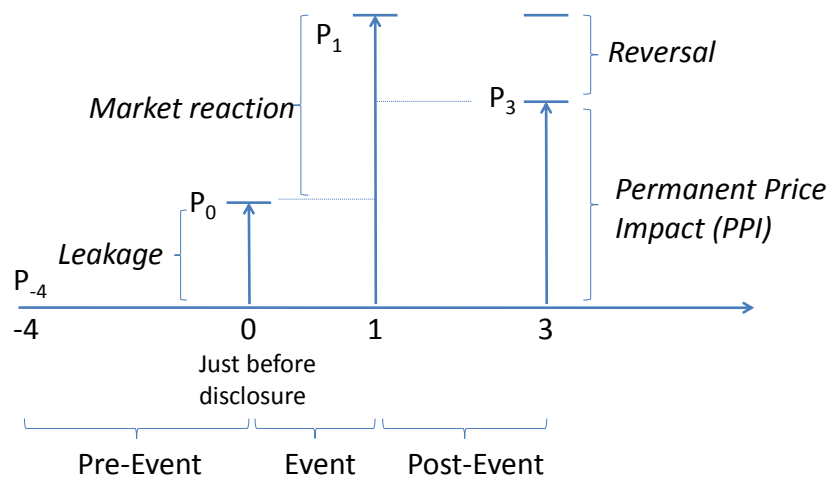


Figure 1a. Decomposition method for the disclosure event.

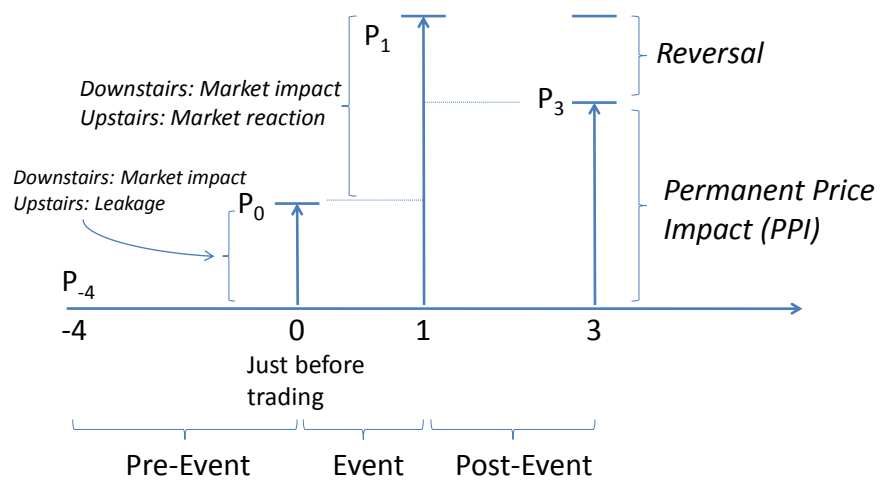


Figure 1b: Decomposition method for the trading event.

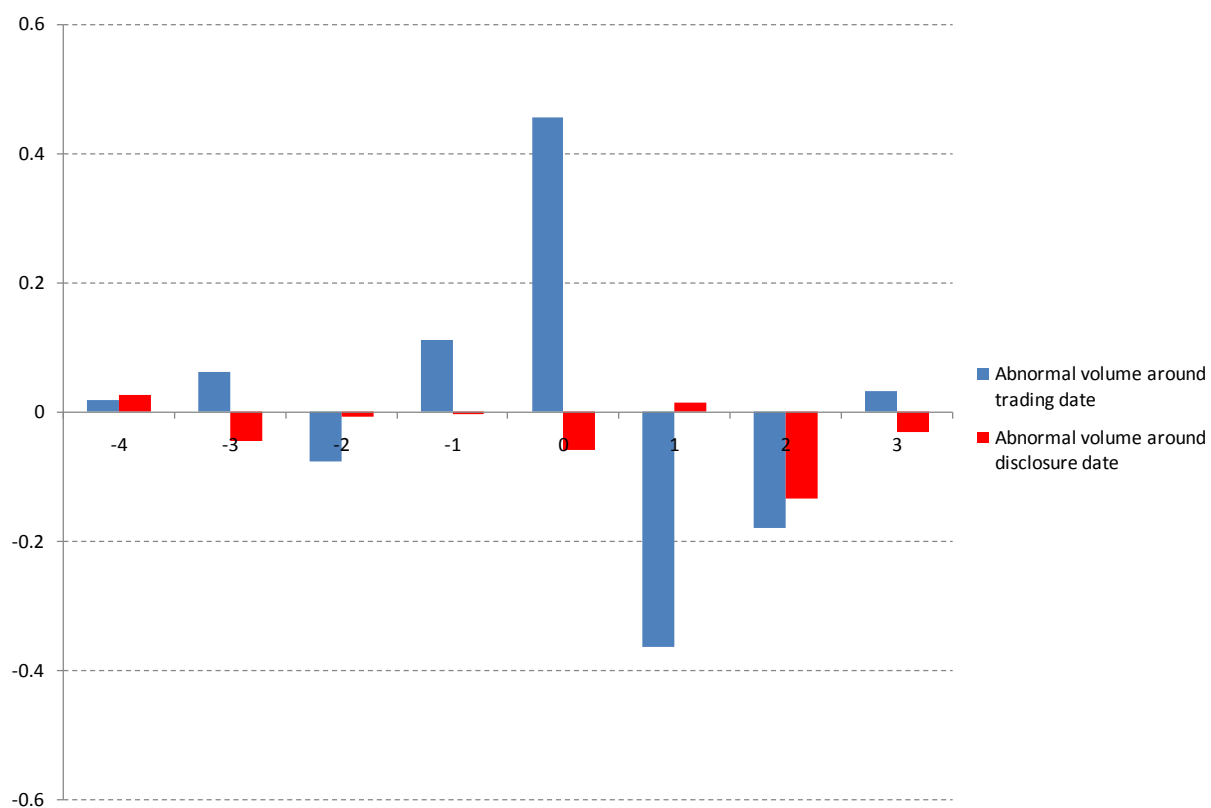


Figure 2a. Abnormal volumes around trading date and disclosure date.

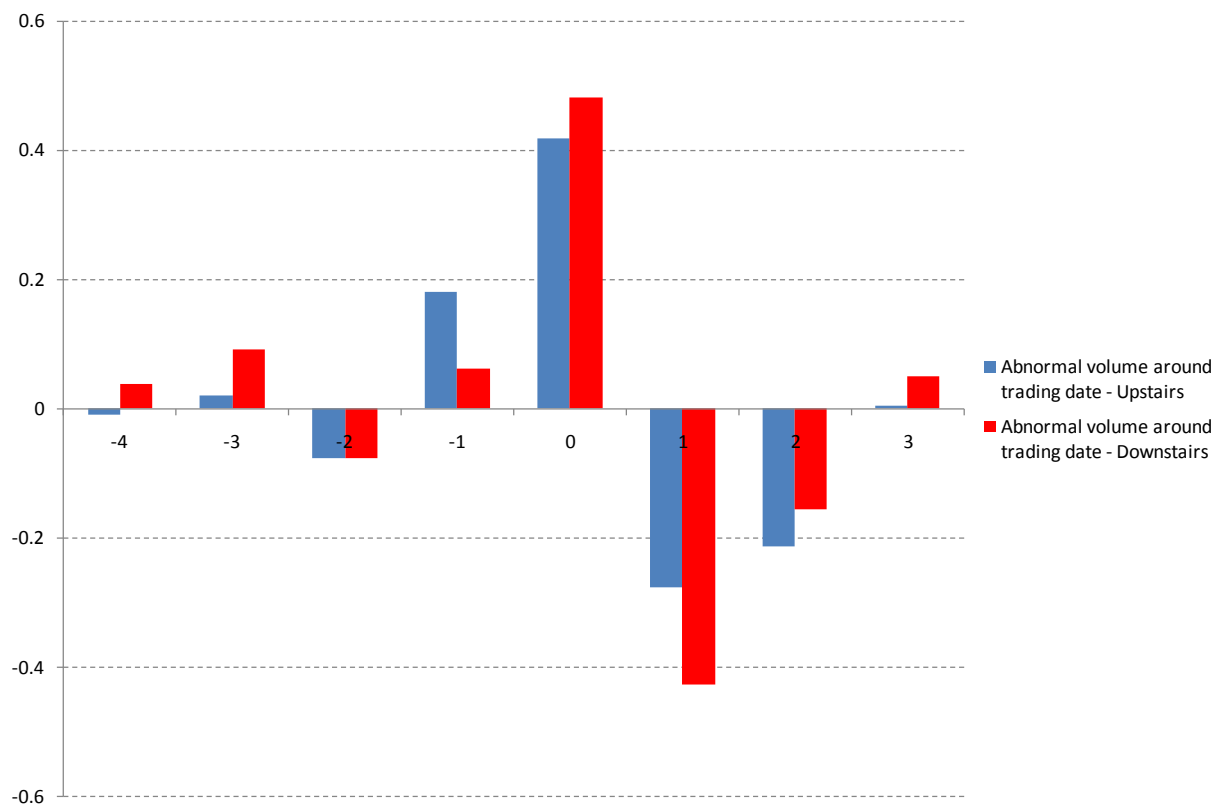


Figure 2b: Abnormal volume around trading date for upstairs and downstairs trades.

Table 1. Previous Studies on Hedge Fund Interventions

The table reports short-term value effects reported in recent studies on hedge funds interventions.

Author(s)	Metric	Event window	
		Narrow	Large
<i>Boyson and Mooradian (2011)</i>		(0, +2)	(-25, +25)
	Average CARs	2.0%	8.1%
<i>Brav et al. (2008)</i>		(0,1)	(-20, +20)
	Average CARs	2.0%	7.2%
	Median CARs		4.6%
<i>Becht et al. (2010)</i>			(-20, +20)
	Average CARs		4.4%
<i>Clifford (2008)</i>		(-2, +2)	
Active hedge funds	Average CARs	3.4%	
	Median CARs	1.7%	
Passive hedge funds	Average CARs	1.6%	
	Median CARs	0.5%	
<i>Greenwood and Schor (2009)</i>		(0)	(-10, +5)
	Average CARs	2.4%	3.6%
<i>Klein and Zur (2009)</i>			(-30, +30)
	Average CARs		7.2%
	Median CARs		5.4%
<i>Mietzner and Schweizer (2011)</i>			(-20, +20)
	Average CARs		4.5%
<i>Weber and Zimmermann (2011)</i>		(-2, +2)	
Transaction date	Average CARs	0.1%	
Announcement date	Average CARs	1.2%	
Publication date	Average CARs	0.6%	

Table 2. Characteristics of the sample

This table presents descriptive statistics of the sample. Panel A reports the number of observations and the average stake acquired by year for both the full sample (161 observations) and the uncontaminated sample (126 observations). Panel B lists the hedge fund that acquired equity stake above 2% in Italian listed companies.

	Full Sample		Uncontaminated Sample	
	# Obs	% Acquired	# Obs	% Acquired
Panel A: Year				
2000	4	4.67%	3	4.62%
2001	11	3.34%	8	3.48%
2002	19	2.34%	13	2.31%
2003	14	2.41%	11	2.40%
2004	14	2.38%	13	2.40%
2005	27	2.80%	20	2.60%
2006	38	2.90%	29	2.48%
2007	34	4.34%	29	4.66%
Total	161	3.11%	126	3.08%
Panel B: Hedge Fund				
Absolute Capital Mgmt Holdings LTD	5	2.06%	5	2.06%
Amber Capital LP	23	3.02%	16	2.21%
Arnhold & S Bleichroeder Advisors	3	2.09%	3	2.09%
Asset Value Investors Ltd	2	2.54%	2	2.54%
Atticus Management Ltd	1	2.02%		
Audley Capital Advisors LLP	1	67.28%	1	67.28%
Boussard & Gavaudan Gestion	1	2.27%	1	2.27%
Centaurus Capital	11	2.53%	9	2.63%
Cheyne Capital Management Limited	1	2.03%		
Cycladic Capital Management LTD	2	3.08%	2	3.08%
Electra Qmc Euro	1	2.24%	1	2.24%
Egerton Capital LP	19	2.35%	14	2.30%
European Small Cap Ltd	1	2.01%	1	2.01%
GLG Partners LP	6	3.09%	5	3.31%
Gartmore Investment Management	14	2.17%	14	2.17%
Harris Associates LP	8	2.07%	7	2.06%
Henderson Investors Ltd	6	3.29%	4	3.11%
JOHM Alternative investment LLD	2	2.19%	2	2.19%
K Capital Partners LLC	5	2.42%	2	2.29%
L.P. Hedge Fund LDT-Global	4	2.60%	4	2.60%
Lansdowne Capital Ltd	6	2.84%	3	3.30%
Laxey Partners Ltd	1	2.63%	1	2.63%
Leonardo Capital Fund SPC limited	11	2.24%	9	2.26%
M2 capital Management LP	1	2.65%	1	2.65%
Oxburgh Partners LLD	1	3.03%		
Parvus European Abs. Opp. Master Fund	1	4.57%	1	4.57%
Pendragon Capital LLP	2	2.12%	2	2.12%
Perry Capital Corp	1	3.75%		
Phoenix Asset Mgmt Partners	1	4.44%		
Sac Capital Advisors	2	6.32%	1	4.46%
Sal. Oppenheim Altern. Inv. GMBH	1	4.54%	1	4.54%
Silchester Intl Investor Ltd	3	6.19%	3	6.19%
Stark Offshore Master Fund	2	2.10%	2	2.10%
The Antares European Fund Limited	2	2.44%	1	2.55%
Trafalgar Asset Managers Ltd	2	3.26%	1	4.04%
TT international	1	2.37%		
Wellington Management Co LLP	6	2.91%	6	2.91%
York Global Finance II Sarl	1	2.18%	1	2.18%

Table 3. Trading and Disclosure Effects

This table reports the abnormal returns around the trading (Panel A) and disclosure (Panel B) dates for both the full sample and the uncontaminated sample. Four event windows are analyzed: the overall price change (-4, +3); pre-event (-4, -1); event (0, +1); post-event (+2, +3). The symbols ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

		Price Change Decomposition			
	# Obs	Total Price Change	Pre-Event	Event	Post-Event
Event Window →		(-4, 3)	(-4, -1)	(0, 1)	(2, 3)
Panel A: Trading					
A1. Full Sample					
Mean	161	1.76%***	0.88%***	0.87%***	0.01%
Median	161	0.71%***	-0.13%	0.34%**	0.05%
% of Total Price Change		100%	50%	50%	0%
A2. Uncontaminated Sample					
Mean	126	1.37%***	0.60%***	0.76%*	0.02%
Median	126	0.64%**	-0.21%	0.36%**	-0.01%
% of Total Price Change		100%	44%	55%	1%
Panel B: Disclosure					
B1. Full Sample					
Mean	161	0.93%**	0.17%	0.55%***	0.21%
Median	161	0.44%**	-0.20%	0.30%*	0.04%
% of Total Price Change		100%	18%	59%	22%
B2. Uncontaminated Sample					
Mean	126	0.44%	0.01%	0.09%	0.34%
Median	126	0.33%	-0.15%	0.07%	0.19%*
% of Total Price Change		100%	2%	20%	78%

Table 4. Trading Effects: Downstairs Market vs. Upstairs Market

This table reports the abnormal returns in the downstairs market (Panel A) and in the upstairs market (Panel B) around the trading dates for both the full sample and the uncontaminated sample. Four event windows are analyzed: the overall price change (-4, +3); pre-event (-4, -1); event (0, +1); post-event (+2, +3). The symbols ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	# Obs	Total Price Change	Price Change Decomposition		
			Pre-Event	Event	Post-Event
Event Window →		(-4, 3)	(-4, -1)	(0, 1)	(2, 3)
Panel A: Downstairs Market					
A1. Full Sample					
Mean	93	2.29%***	1.61%***	0.68%***	-0.01%
Median	93	0.58%	0.10%	0.20%**	-0.07%
% of Total Price Change		100%	70%	30%	0%
A2. Uncontaminated Sample					
Mean	74	1.57%**	0.99%***	0.50%***	0.08%
Median	74	0.48%	-0.15%	0.19%**	-0.01%
% of Total Price Change		100%	63%	32%	5%
Panel B: Upstairs Market					
B1. Full Sample					
Mean	68	1.04%*	-0.11%	1.13%***	0.03%
Median	68	0.93%**	-0.34%*	0.37%	0.16%
% of Total Price Change		100%	-11%	108%	3%
B2. Uncontaminated Sample					
Mean	52	1.08%*	0.04%	1.12%***	-0.07%
Median	52	1.16%*	-0.38%	0.47%	-0.01%
% of Total Price Change		100%	3%	103%	-7%

Table 5. Trading Effects: Buy Blocks vs. Sell Blocks

This table reports the abnormal returns for buy blocks (Panel A) and sell blocks (Panel B) around the trading dates for both the full sample and the uncontaminated sample. Four event windows are analyzed: the overall price change (-4, +3); pre-event (-4, -1); event (0, +1); post-event (+2, +3). The symbols ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

			Price Change Decomposition		
	# Obs	Total Price Change	Pre-Event	Event	Post-Event
Event Window →		(-4, 3)	(-4, -1)	(0, 1)	(2, 3)
Panel A: Buy Blocks					
A1. Full Sample					
Mean	26	1.93%**	-0.03%	1.88%***	0.08%
Median	26	-0.86%	-0.52%	0.36%**	-0.21%
% of Total Price Change		100%	-1%	97%	4%
A2. Uncontaminated Sample					
Mean	17	2.44%**	-0.13%	2.25%***	0.33%
Median	17	-0.48%	-0.65%	1.12%**	-0.10%
% of Total Price Change		100%	-5%	92%	14%
Panel B: Sell Blocks					
B1. Full Sample					
Mean	42	0.49%	-0.17%	0.66%	-0.01%
Median	42	1.32%	-0.24%	0.39%	0.24%
% of Total Price Change		100%	-34%	135%	-1%
B2. Uncontaminated Sample					
Mean	35	0.42%	0.12%	0.57%	-0.27%
Median	35	1.20%	-0.22%	0.37%	0.03%
% of Total Price Change		100%	29%	136%	-65%

Table 6. Multivariate Regressions

This table reports the estimates of OLS regression models where the dependent variable is the CAR in the event window (-4, +3) in Panel A, the CAR in event window (-4, -1) in Panel B, and the CAR in the event window (0, 1) in Panel C on a constant, the percentage of a target's equity disclosed, a dummy for upstairs trades (Block) and a dummy for buy blocks (Buy Block). Robust standard errors are in parenthesis. The symbols ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>Trading Date</i>			<i>Disclosure Date</i>		
	I	II	III	IV	V	VI
Panel A: CAR (-4, 3)						
Constant	0.0146** [0.0059]	0.0198** [0.0083]	0.0199** [0.0083]	0.0084* [0.0044]	0.0082 [0.0065]	0.0084 [0.0065]
% of equity	0.0976* [0.0513]	0.0919 [0.0563]	0.0873* [0.0516]	0.0292 [0.0549]	0.0294 [0.0553]	0.0232 [0.0494]
Block		-0.0119 [0.0110]	-0.0172 [0.0117]		0.0004 [0.0084]	-0.0066 [0.0088]
Buy Block			0.0136 [0.0169]			0.0183 [0.0121]
Adjusted R ²	-0.0006	0.0003	-0.0022	-0.0054	-0.0118	-0.0068
N. Obs.	161	161	161	161	161	161
Panel B: CAR (-4, -1)						
Constant	0.0145** [0.0058]	0.0198** [0.0087]	0.0200** [0.0087]	0.0071* [0.0040]	0.006 [0.0057]	0.0061 [0.0057]
% of equity	0.0988** [0.0387]	0.0930** [0.0437]	0.0887** [0.0398]	0.0045 [0.0582]	0.0057 [0.0585]	0.0028 [0.0549]
Block		-0.0123 [0.0106]	-0.0172 [0.0117]		0.0026 [0.0075]	-0.0007 [0.0080]
Buy Block			0.0128 [0.0138]			0.0087 [0.0112]
Adjusted R ²	-0.0003	0.0011	-0.0019	-0.0063	-0.0119	-0.0149
N. Obs.	161	161	161	161	161	161
Panel C: CAR (0, 1)						
Constant	0.0044 [0.0030]	0.0021 [0.0044]	0.0022 [0.0044]	0.0057** [0.0022]	0.0072** [0.0034]	0.0073** [0.0034]
% of equity	0.1402*** [0.0388]	0.1427*** [0.0380]	0.1390*** [0.0336]	-0.0047 [0.0091]	-0.0063 [0.0093]	-0.0096 [0.0109]
Block		0.0052 [0.0057]	0.0011 [0.0062]		-0.0035 [0.0039]	-0.0072* [0.0041]
Buy Block			0.0109 [0.0081]			0.0098** [0.0045]
Adjusted R ²	0.0356	0.0344	0.037	-0.0062	-0.0082	-0.0004
N. Obs.	161	161	161	161	161	161

Table 7. Volume Event Study: Trading and Disclosure Effects

This table reports the estimates of the daily abnormal trading volumes in the eight-day period (-4, +3) around trading and disclosure dates for the full sample (Panel A) and the uncontaminated sample (Panel B). To perform the event study on the firm's trading volume, we follow a market model approach like Sanders and Zdanovic (1992). As in Ajinkya and Jain (1989), we do not perform the event study on the raw trading volume, which is highly non-normal, but on the transformed variable $v_{it} = (1 + \ln \text{volume}_{it})$. The trading volume of the local market index is used in the market model regressions. As in the stock returns, we estimate market model parameters over the period (-240, -41). The symbols ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Trading		Disclosure		N
	Mean	Median	Mean	Median	
Panel A: Full Sample					
Day -4	0.0184	0.0300	0.0257	-0.0451	161
Day -3	0.0620	0.0221	-0.0451	-0.0133	161
Day -2	-0.0758	-0.0388	-0.0065	-0.0358	161
Day -1	0.1121	0.1120**	-0.0037	-0.1178	161
Day 0	0.4553***	0.2079***	-0.0584	-0.0870	161
Day 1	-0.3633***	-0.4016***	0.0145	-0.0343	161
Day 2	-0.1795***	-0.1785***	-0.1334*	-0.0619*	161
Day 3	0.0320	-0.0641	-0.0302	0.0581	161
Panel B: Uncontaminated Sample					
Day -4	0.0312	0.0012	-0.0029	-0.1153	126
Day -3	0.0728	0.0216	-0.0668	-0.0293	126
Day -2	-0.1478	-0.0567	0.0302	-0.0858	126
Day -1	0.1137	0.0459	-0.0102	-0.1253	126
Day 0	0.3945***	0.1992***	-0.0704	-0.1053	126
Day 1	-0.3787***	-0.3448***	0.0011	-0.0490	126
Day 2	-0.1246	-0.1467	-0.1556**	-0.0617*	126
Day 3	0.0776	-0.0582	0.0254	0.0800	126

Author Biographies

Ettore Croci is Assistant Professor at Università Cattolica del Sacro Cuore, Milan, Italy. His research interests include empirical corporate finance, in particular M&A, corporate governance, and shareholder activism. His publications have appeared in journals such as *Journal of Corporate Finance*, *Journal of Banking & Finance*, *European Financial Management*, and *European Journal of Finance*.

Giovanni Petrella is Associate Professor at Università Cattolica del Sacro Cuore, Milan, Italy. His research interests include empirical market microstructure, derivatives, and portfolio management. His publications have appeared in journals such as *Journal of Banking & Finance*, *European Financial Management*, *Journal of Futures Markets*, *Journal of Trading*, and *Quantitative Finance*.